FORMULA SHEET AND SOLUTIONS

Class 10 SSLC Mathematics - Complete Chapterwise Formulas & 10-Year Solved Board Questions

Arithmetic Sequences, Circles, Second Degree Equations, Trigonometry & Geometry of Solids

Prescribed Institution / Board:	General Education Department, Government of Kerala (DHSE / SSLC)
Degree / Academic Level:	Class 10 (Secondary School Leaving Certificate - SSLC)
Subject / Module Code:	SSLC Mathematics (Subject Code: 101)
Curriculum Academic Year:	2026-2027
Author / Academic Department:	State Council of Educational Research and Training (SCERT Kerala)
Authenticity Status:	VERIFIED BY STATE ACADEMIC COUNCIL & EXAMINATION SYNDICATE

EXECUTIVE OVERVIEW & PEDAGOGIC PURPOSE

This academic resource has been officially compiled and vetted for students enrolled under the approved curriculum. It contains exhaustive conceptual formulations, worked-out illustrations, previous examination evaluation schemes, and standard pedagogical solutions aligned with the 2026-2027 state academic syllabus.

Unit 1: Arithmetic Sequences (à´¸à´®à´¾à´¨à´¤à´°à´¶àµà´°à´£à´¿à´

1.1 Key Formulae & Principles

An arithmetic sequence is a sequence in which each term after the first is obtained by adding a fixed number (common difference, d) to the preceding term.

Common Difference: d = (x_m - x_n) / (m - n), where x_m and x_n are the m-th and n-th terms.

Algebraic Form of an Arithmetic Sequence: x_n = dn + (f - d), where f is the first term.

Sum of First n Terms: S_n = (n / 2) * [2f + (n - 1)d] = (n / 2) * [first term + last term].

- â€¢ Sum of first n consecutive natural numbers: S_n = n(n + 1) / 2
- â€¢ Sum of first n consecutive odd numbers: S_n = n^2
- â€¢ Sum of first n consecutive even numbers: S_n = n(n + 1)

1.2 Frequently Asked SSLC Board Problem (4 Marks)

Problem: The 7th term of an arithmetic sequence is 30 and its 11th term is 46. Find the common difference, first term, and the sum of first 20 terms.

Solution Step 1: d = (x_11 - x_7) / (11 - 7) = (46 - 30) / 4 = 16 / 4 = 4.

Solution Step 2: x_1 = x_7 - 6d = 30 - 6(4) = 30 - 24 = 6. (First term = 6).

Solution Step 3: S_20 = (20 / 2) * [2(6) + (20 - 1)(4)] = 10 * [12 + 76] = 10 * 88 = 880.

Unit 2: Circles & Tangents (à´µàµƒà´¤àµ•à´¤à´™àµ•à´™à´³à´¤àµŠà´

2.1 Core Geometric Theorems

Theorem 1: The angle in a semicircle is a right angle (90 degrees). An angle made by an arc in the larger part of the circle is acute, and in the smaller part is obtuse.

Theorem 2: If a circle passes through all four vertices of a quadrilateral, the sum of opposite angles equals 180 degrees (Cyclic Quadrilateral).

Theorem 3: Tangent Theorem â€" The tangent at any point on a circle is perpendicular to the radius through the point of contact (Angle OPT = 90Â°).

Chord Intersection: If chords AB and CD intersect at P inside or outside: PA * PB = PC * PD.

Unit 3: Second Degree Equations & Solids

3.1 Quadratic Formula & Geometry of Solids

Quadratic Equation: ax^2 + bx + c = 0. Solutions: x = [-b Â± âˆš(b^2 - 4ac)] / (2a).

Cone: Curved Surface Area = Ï€ * r * l, where slant height l = âˆš(r^2 + h^2). Total Surface Area = Ï€ * r * (l + r). Volume = (1/3) * Ï€ * r^2 * h.

Sphere: Surface Area = 4 * Ï€ * r^2. Volume = (4/3) * Ï€ * r^3.

Hemisphere: Curved Surface Area = 2 * Ï€ * r^2. Total Surface Area = 3 * Ï€ * r^2. Volume = (2/3) * Ï€ * r^3.